

On the Divisibility of Linear Codes and Arcs

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Abstract

We prove several divisibility results for codes over non-prime fields. They can be viewed as a generalization of Ward's classical divisibility theorem for Griesmer codes over prime fields.

Key Words : divisible codes, divisible arcs, Griesmer codes, extendible codes, extendible arc.

1 Introduction

Let $\mathbb{F}_q$ be the finite field with q elements and let $\mathbb{F}_q^n$ denote the vector space of all ordered n -tuples over $\mathbb{F}_q$. A linear code of length n and dimension k is a k -dimensional subspace of $\mathbb{F}_q^n$. The minimum (Hamming) distance of a linear code C is the smallest of the distances between different codewords of C . A linear code over $\mathbb{F}_q$ of length n , dimension k , and minimum distance d is referred to as an $[n, k, d]_q$ -code. A linear code is said to be divisible by Δ if the weight of every codeword is divisible by Δ . It is known that if Δ is relatively prime to the field size q , then a code of full length (no coordinate identically zero) divisible by Δ is a Δ -fold replicated code, a code obtained from a shorter code by repeating each coordinate Δ times.

We denote by $n_q(k, d)$ the minimal length of a linear code of fixed dimension k and fixed minimum distance d over a field with q elements. A

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lower bound on $n_q(k, d)$ is given by:

$$n_q(k, d) \geq g_q(k, d) := \sum_{i=0}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil. \quad (1)$$

The bound (1) was proved originally by Griesmer [3] for binary codes and generalized for q -ary codes by Solomon and Stiffler [6]. An $[n, k, d]_q$ -code with $n = g_q(k, d)$ is called a Griesmer code. The Griesmer codes exhibit strong divisibility properties.

Dodunekov and Manev proved in [2] that for a binary Griesmer code the highest power of the code is also a divisor of the code itself. This result was extended to linear codes over arbitrary prime fields [7].

Theorem 1. (*H. N. Ward*) *Let C be a Griesmer code over $\mathbb{F}_p$, p a prime. Then if p^e divides the minimum weight of C , p^e is a divisor of the code.*

Theorem 1 imposes restrictions on the weight enumerator of a hypothetical Griesmer code. Sometimes they are so strong that a particular set of parameters turns out to be impossible. For Griesmer codes over non-prime fields we have only weaker results. The following theorems were proved by H. N. Ward in [8].

Theorem 2. (*H. N. Ward*) *Let C be a $[g_q(k, d), k, d]_q$ Griesmer code, and suppose that q divides d . Then C is divisible by p , the prime dividing q .*

Theorem 3. (*H. N. Ward*) *Let C be a $[g_4(k, d), k, d]_4$ -code for which 2^e divides d with $e \geq 2$. Then 2^{e-1} is a divisor of C .*

The following conjecture by Ward stated in [8] is also worth mentioning.

Conjecture 1. (*H. N. Ward*) *Let C be a $[g_q(k, d), k, d]_q$ Griesmer code where q is a power of the prime p . Suppose that p^e divides d with $p^e > q$. Then C is divisible by p^{e+1}/d .*

It turns out that under certain additional conditions non-Griesmer codes are also divisible. In [1] it was proved that if the number of 2-points in an $(q^2 + q + 2, q + 2)$ -arc is larger than $(q - 1)p^{t-1}$ then the multiplicity of each line is congruent to $q^2 + q + 2 \pmod{p^t}$.

In this paper we present some new divisibility results for codes over non-prime fields. We also give a condition on a non-Griesmer code that forces it to be divisible.

2 Linear codes and arcs

There exists a connection of the linear codes with the multisets of points in finite projective geometries. Let $\mathcal{P}$ be the set of points of $\text{PG}(k-1, q)$. Every

mapping $\mathcal{K} : \mathcal{P} \rightarrow \mathbb{N}_0$ is called a multiset in $\text{PG}(k-1, q)$. This mapping is extended to the subsets S of $\mathcal{P}$ by $\mathcal{K}(S) = \sum_{P \in S} \mathcal{K}(P)$. The integer $\mathcal{K}(P)$, $P \in \mathcal{P}$, is called the multiplicity of the point P , and $n = \sum_{P \in \mathcal{P}} \mathcal{K}(P)$ is the cardinality of the multiset $\mathcal{K}$. A multiset is said to be divisible with divisor Δ if $\mathcal{K}(H) \equiv n \pmod{\Delta}$ for every hyperplane H .

A multiset $\mathcal{K}$ in $\text{PG}(r, q)$ is called an (n, w) -arc if its cardinality is n , $\mathcal{K}(H) \leq w$ for every hyperplane H , and there exists a hyperplane H_0 with $\mathcal{K}(H_0) = w$.

It is known that there exists a one-to-one correspondence between the isomorphism classes of the linear $[n, k]_q$ -codes of full length (no coordinate is identically zero in all codewords) and the classes of projectively equivalent spanning multisets in $\text{PG}(k-1, q)$, where $k \geq 2$. The correspondence can be described as follows. Let C be a linear code of full length with parameters $[n, k, d]_q$, and let $G = (g_1^T \cdots g_n^T)$, $g_i \in \mathbb{F}_q^k$, be a generator matrix of C . The columns g_i^T are considered as the homogeneous coordinates of points in $\text{PG}(k-1, q)$. In this way, the generator matrix G is associated with an ordered n -tuple of points $(P_1, \dots, P_n)$ in $\text{PG}(k-1, q)$. This n -tuple defines a multiset $\mathcal{K}$ by $\mathcal{K}(P) := |\{i | P_i = P\}|$.

An arc in $\text{PG}(k-1, q)$ associated with a linear $[n, k, d]_q$ -code C has parameters $(n, n-d)$. A codeword $c \in C$ of Hamming weight $w(c)$ corresponds to a hyperplane in H of multiplicity $n - w(c)$. The code C has a divisor Δ if and only if the associated arc is divisible with the same divisor.

An arc associated with a Griesmer code is called a Griesmer arc. Let C be an $[n = g_q(k, d), k, d]_q$ -code and let d be written as:

$$d = sq^{k-1} - \lambda_{k-2}q^{k-2} - \dots - \lambda_1q - \lambda_0, \quad (2)$$

where $0 \leq \lambda_i \leq q-1$. It is easily checked that

$$n = g_q(k, d) = sv_k - \lambda_{k-2}v_{k-1} - \dots - \lambda_1v_2 - \lambda_0v_1, \quad (3)$$

where $v_i = (q^i - 1)/(q - 1)$. For the highest multiplicity $w_q(k, d)$ of a hyperplane with respect to the (n, w) -arc $\mathcal{K}$ associated with C it holds

$$w_q(k, d) = sv_{k-1} - \lambda_{k-2}v_{k-2} - \dots - \lambda_1v_1. \quad (4)$$

Let $\mathcal{K}$ be a $(g_q(k, d) + t, w_q(k, d) + t)$ -arc in $\text{PG}(k-1, q)$, where d is given by (2). The following lemma known as a folklore.

Lemma 1. *Let $d = sq^{k-1} - \lambda_{k-2}q^{k-2} - \dots - \lambda_0$ and let $\mathcal{K}$ be a $(g_q(k, d) + t, w_q(k, d) + t)$ -arc in $\text{PG}(k-1, q)$. Then for every j -dimensional subspace S in $\text{PG}(k-1, q)$*

$$\mathcal{K}(S) \leq t + \sum_{i=k-1-j}^{k-1} \left\lceil \frac{d}{q^i} \right\rceil.$$

The following formula is a direct consequence of Lemma 1 for a Griesmer arc $\mathcal{K}$ with parameters given by (3) and (4). Denote by w_j , $j = 0, \dots, k-1$, the maximal multiplicity of a subspace of dimension j with respect to $\mathcal{K}$. Then

$$w_j = sv_{j+1} - \lambda_{k-2}v_j - \dots - \lambda_{k-1-j}v_1. \quad (5)$$

Note that if q^e divides $d = n - w$ then $\lambda_0 = \dots = \lambda_{e-1} = 0$.

In what follows we shall need multisets induced by a projection from a subspace. Let $\mathcal{K}$ be a multiset in $\text{PG}(k-1, q)$. Fix an i -dimensional subspace δ in $\text{PG}(k-1, q)$, with $\mathcal{K}(\delta) = t$. Let further π be a complementary j -dimensional subspace in $\text{PG}(k-1, q)$, i.e. $i + j = k-2$ and $\delta \cap \pi = \emptyset$. Define the projection $\varphi = \varphi_{\delta, \pi}$ from δ onto π by

$$\varphi: \begin{cases} \mathcal{P} \setminus \delta & \rightarrow \pi \\ Q & \rightarrow \pi \cap \langle \delta, Q \rangle. \end{cases} \quad (6)$$

As before, $\mathcal{P}$ denotes the set of all points of $\text{PG}(k-1, q)$. Note that $\varphi_{\delta, \pi}$ maps $(i+s)$ -subspaces containing δ into $(s-1)$ -subspaces in π . Denote by $\mathcal{P}'$ the set of all points in π . We define the induced multiset $\mathcal{K}^\varphi : \mathcal{P}' \rightarrow \mathbb{N}_0$ by

$$\mathcal{K}^\varphi(Q) = \sum_{P \in \mathcal{P} \setminus \delta : \varphi(P) = Q} \mathcal{K}(P).$$

It is clear that for every subspace S in $\text{PG}(k-1, q)$ that contains δ , it holds $\mathcal{K}^\varphi(\varphi(S)) = \mathcal{K}(S) - t$.

3 A Divisibility Theorem

Let $\mathcal{K}$ be an arc of cardinality n in $\text{PG}(k-1, q)$. Let $e \geq 1$ be an integer. A subspace S of dimension $k-3$ (co-dimension 2) is said to have the property (D) if every hyperplane H through S satisfies $\mathcal{K}(H) \equiv n \pmod{q^e}$. Let us note that we do not require $\mathcal{K}$ to be necessarily a Griesmer arc. For Griesmer arcs with divisor q^e all subspaces of maximal multiplicity of codimension 2 have property (D) .

Theorem 4. *Let $\mathcal{K}$ be an (n, w) -arc in $\text{PG}(k-1, q)$, and let $e \geq 1$ be an integer. If there exists a subspace δ of codimension 3 with the properties*

- (1) $\mathcal{K}(\delta) \equiv n \pmod{q^{e-2}}$ if $e \geq 2$;
- (2) for every subspace S of codimension 2 through δ it holds $\mathcal{K}(S) \equiv n \pmod{q^{e-1}}$;
- (3) there exist at least q subspaces of codimension 2 through δ with the property (D) .

then $\mathcal{K}(H) \equiv n \pmod{pq^{e-1}}$ for every hyperplane H containing δ .

Proof. Consider a projection φ from δ onto some plane π with $\delta \cap \pi = \emptyset$. Set $n' = n - \mathcal{K}(\delta)$, $w' = w - \mathcal{K}(\delta)$. Clearly $\mathcal{K}^\varphi$ is an (n', w') -arc in $\pi \cong \text{PG}(2, q)$.

Define a new arc $\mathcal{F}$ by

$$\mathcal{F}(P) = \frac{n' - \mathcal{K}^\varphi(\langle \delta, P \rangle)}{q^{e-1}},$$

for every point P on π . Set $N := |\mathcal{F}|$.

Let $P \in \pi$ be the image of a hyperline $S \supset \delta$ with the property (D). For every line L through P we have

$$\begin{aligned} \mathcal{F}(L) &= \sum_{P \in L} \mathcal{F}(P) \\ &= \frac{1}{q^{e-1}} \sum_{P \in L} (n' - \mathcal{K}^\varphi(\langle \delta, P \rangle)) \\ &= \frac{1}{q^{e-1}} ((q+1)n' - \mathcal{K}(H) + \mathcal{K}(\delta)). \end{aligned}$$

On the other hand,

$$\begin{aligned} N &= \sum_{P \in \pi} \mathcal{F}(P) \\ &= \frac{1}{q^{e-1}} \sum_{P \in \pi} (n' - \mathcal{K}^\varphi(\langle \delta, P \rangle)) \\ &= \frac{1}{q^{e-1}} ((q^2 + q + 1)n' - (n - \mathcal{K}(\delta))) \\ &= \frac{(q^2 + q)n'}{q^{e-1}}. \end{aligned}$$

Now

$$\begin{aligned} N - \mathcal{F}(L) &= \frac{1}{q^{e-1}} ((q^2 + q)n' - (q+1)n' + \mathcal{K}(H) - \mathcal{K}(\delta)) \\ &= \frac{1}{q^{e-1}} (q^2 n' - n + \mathcal{K}(H)). \end{aligned}$$

Since $n' \equiv 0 \pmod{q^{e-2}}$ we obtain that $\mathcal{F}(L) \equiv N \pmod{q}$ for every line L that is the image of a hyperplane H with $\mathcal{K}(H) \equiv n \pmod{q^e}$.

Now let H_0 be an arbitrary hyperplane through δ and set $L_\infty = \varphi(H_0)$. We identify the points from π with the elements of $\mathbb{F}_{q^2}$. Two lines $\langle x, y \rangle$ and $\langle x', y' \rangle$, $x, y, x', y' \in \mathbb{F}_{q^2}$, are parallel iff $(x - y)^{q-1} = (x' - y')^{q-1}$. Hence the points on L_∞ can be identified with the $(q+1)$ -st roots of unity in $\mathbb{F}_{q^2}$. We denote this subgroup of $\mathbb{F}_{q^2}^*$ by $\mathcal{U}_{q+1}$.

Without loss of generality we can assume that for every subspace of codimension 2 through δ with the property (D): $\varphi(\Delta) \neq 0$ (Here 0 is the

zero in $\mathbb{F}_{q^2}$.) Now consider the locator polynomial

$$\begin{aligned} F(X, Y) &= \prod_{s \in \mathbb{F}_{q^2}} (1 - (1 - sX)^{q-1}Y)^{\mathcal{F}(s)} \prod_{\zeta \in \mathcal{U}_{q+1}} (1 - \zeta X^{p-1}Y)^{\mathcal{F}(\zeta)} \\ &= \sum_{i=0}^N F_i(X) Y^i. \end{aligned}$$

Note that $\deg F_i \leq i(q-1)$.

Let $t \in \mathbb{F}_{q^2}$ and let t be the image under φ of a subspace of codimension 2 through δ with the property (D). If $s \neq t$ we have that $(1 - st^{-1})^{p-1} = ut^{-1}$ for some $u \in \mathcal{U}_{q+1}$. Here u is associated with the point $(t-s)^{q-1}$ which is the common point of L_∞ and the line $\langle s, t \rangle$. Denote by L_ζ the line through t and ζ then

$$F(t^{-1}, Y) = \prod_{\zeta \in \mathcal{U}_{q+1}} (1 - \zeta Y)^{\mathcal{F}(L_\zeta - \mathcal{F}(t))}. \quad (7)$$

The numbers $\mathcal{F}(L_\zeta)$, $\zeta \in \mathcal{U}_{q+1}$, are equal modulo q and hence we set $\mathcal{F}(L_\zeta) - \mathcal{F}(t) = a + b_\zeta q$, where $0 \leq a < q$, and b_ζ . From (7) we get

$$\begin{aligned} F(t^{-1}, Y) &= \prod_{\zeta \in \mathcal{U}_{p+1}} (1 - \zeta Y)^{a+b_\zeta q} \\ &= (1 - Y)^{a(q+1)} \prod_{\zeta \in \mathcal{U}_{p+1}} (1 - \zeta Y)^{b_\zeta q} \\ &= (1 - Y)^{a(q+1)} \prod_{\zeta \in \mathcal{U}_{p+1}} (1 - (\zeta Y)^{b_\zeta q}). \end{aligned}$$

This implies that $F_i(t^{-1}) = 0$ for $i = 1, \dots, q-1$. Since $\deg F_1 \leq q-1$ and since there are at least q elements $t \in \mathbb{F}_{q^2}$ (points in $\pi \setminus L_\infty$) we get that $F_1(X) \equiv 0$.

On the other hand, we have that

$$\begin{aligned} F(0, Y) &= (1 - Y)^{N - \mathcal{F}(L_\infty)} \\ &= 1 - \binom{N - \mathcal{F}(L_\infty)}{1} Y + \binom{N - \mathcal{F}(L_\infty)}{2} Y^2 - \dots \end{aligned}$$

The coefficient of Y is zero. This implies $\mathcal{F}(L_\infty) \equiv N \pmod{p}$. Plugging in the values for the left and the right side, we get

$$\frac{(q+1)n' - \mathcal{K}(H_0) + \mathcal{K}(\delta)}{q^{e-1}} \equiv \frac{q(q+1)n'}{q^{e-1}} \pmod{p}$$

By (1) we have that $n' \equiv 0 \pmod{q^{e-2}}$. Using also the obvious $n = n' + \mathcal{K}(\delta)$, we get

$$\frac{n - \mathcal{K}(H_0)}{q^{e-1}} \equiv 0 \pmod{p},$$

which in turn implies $\mathcal{K}(H_0) \equiv n \pmod{pq^{e-1}}$. $\square$

Lemma 2. *For a Griesmer (n, w) -arc in $\text{PG}(k-1, q)$, every hyperplane with a point of maximal multiplicity contains a subspace of codimension 3 that satisfies properties (1)–(3).*

Proof. Without loss of generality $e \leq k-2$. Next we proceed by induction on k . For $k=3$, we can take $e=1$. Properties (1) and (2) are automatically satisfied. Now:

$$d = sq^2 - \lambda_1 q, n = sv_3 - \lambda_1 v_2, w_1 = sv_2 - \lambda_1 v_1, w_0 = s.$$

If we assume that there are less than q points of multiplicity s we get

$$sv_3 - \lambda_1 v_2 = n \leq (q-1)s + (q^2+2)(s-1) < sv_3 - \lambda_1 v_2,$$

a contradiction.

Further we proceed by induction. Consider a projection φ from the maximal point and apply the induction hypothesis to the induced arc $\mathcal{K}^\varphi$ which has parameters

$$n' = n - s = (sq - \lambda_{k-2})v_{k-1} - \lambda_{k-3}v_{k-2} - \lambda_e v_{e+1},$$

$$w' = w - s = (sq - \lambda_{k-2})v_{k-2} - \lambda_{k-3}v_{k-3} - \lambda_e v_e,$$

and is therefore a Griesmer arc with $q^e | d' = n' - w'$. $\square$

Theorem 5. *Let C be a $[g_q(k, d), k, d]_q$, $q = p^h$, be a Griesmer code and let d be given by (2) where $s=1$, and suppose that q^e , $e \geq 2$, divides d . Then C is divisible by $p^2 q^{e-2}$.*

Proof. Let $\mathcal{K}$ be the arc associated with C . By Theorem 4 and Lemma 2 every hyperplane H which is not empty satisfies $\mathcal{K}(H) \equiv n \pmod{pq^{e-1}}$.

Now let H be an empty hyperplane. Consider a subspace S in H of codimension 2, and assume that x hyperplanes through S are non-zero hyperplanes. Hence $n \equiv xn \pmod{pq^e}$ and $(x-1)n \equiv 0 \pmod{pq^{e-1}}$. Since $x \leq q$ we get that the highest power of p dividing $x-1$ is $h-1$. Hence C has a divisor $\frac{pq^{e-1}}{p^{h-1}} = p^2 q^{e-2}$. $\square$

For $q=p$, this result implies the classical Ward theorem in the special case of projective codes.

4 Quasidivisible Arcs and Codes

We call an (n, w) -arc $\mathcal{K}$ in $\text{PG}(k-1, q)$ t -quasidivisible with divisor Δ if $\mathcal{K}(H) \equiv n, n+1, \dots, n+t \pmod{\Delta}$. Similarly, a $[n, k, d]_q$ -code C is called t -quasidivisible if $\text{wt}(c) \equiv -t, -(t-1), \dots, 0 \pmod{\Delta}$ for every codeword

$c \in C$. As before, t -quasidivisible arcs are associated with t -quasidivisible codes. For $t = 1$, we say that $\mathcal{K}$ (resp. C) is just quasidivisible.

By Hill-Lizak's extension lemma [4, 5], every quasidivisible arc (resp. quasidivisible code) with divisor q is extendable to a divisible arc (resp. divisible code) with the same divisor.

Let S be a subspace of codimension 2 in $\text{PG}(k-1, q)$ and let $H_0, \dots, H_q$ be the hyperplanes through S . The subspace S is said to have the property (QD) if :

- $\mathcal{K}(H_i) \equiv n+1 \pmod{q}$ for q of the hyperplanes through S ,
- $\mathcal{K}(H_i) \equiv n \pmod{q}$ for exactly one hyperplane H_i .

Using a similar technique as in the previous section, we can prove the following theorems.

Theorem 6. *Let $\mathcal{K}$ be a (n, w) -arc in $\text{PG}(k-1, p)$, p -prime, with $w \equiv n+1 \pmod{p}$. Assume δ is a subspace of codimension 3 with the property that there exist at least $2p-1$ subspaces of codimension 2 through δ having the property (QD) with $q = p$. Then for every hyperplane H through δ it holds*

$$\mathcal{K}(H) \equiv n \text{ or } n+1 \pmod{p}.$$

Theorem 7. *Let $\mathcal{K}$ be a Griesmer (n, w) -arc in $\text{PG}(k-1, p)$, p -prime, with $w \equiv n+1 \pmod{p}$. Let*

$$d = n - w = sp^{k-1} - \lambda_{k-2}p^{k-2} - \dots - \lambda_1p - 1,$$

with $\lambda_i \leq p-2$ for $i = 1, \dots, k-2$. Then $\mathcal{K}$ is 1-quasidivisible and hence extendable to a divisible Griesmer arc with divisor p .

Example 1. *Consider a blocking set $\mathcal{B}$ in $\text{PG}(2, q)$ with parameters $(x(p+1)+1, x)$, $x \leq p-1$. Denote by Λ_i the number of points of multiplicity i . By Theorem 7 (applied for the arc $\mathcal{K} = 2 - \mathcal{B}$) if $\Lambda_0 \geq 2p-1$, $\mathcal{B}$ is reducible to a canonical $(x(p+1), x)$ -blocking set. This is always the case if $x \leq p-2$. For $x = p-1$ irreducible blocking sets are possible.*

For an irreducible $(p^2, p-1)$ -blocking set $\mathcal{B}$ the undecided cases are $p+1 \leq \Lambda_0 \leq 2p-2$. If $\Lambda_0 = p+1$ then $\mathcal{B}$ is the complement of an oval. The classification for $p+2 \leq \Lambda_0 \leq 2p-2$ is unknown. There exist $(p^2, p-1)$ -blocking sets for which the number of 0-points is in this interval. For instance, there exists a $(25, 4)$ -blocking set in $\text{PG}(2, 5)$ with $\Lambda_0 = 8$ ($\Lambda_2 = 2$). It would be of interest to find another examples.

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